PSF

Platform Feature Map

Pakistan Strategic Forum | forumstrategic.com | Website and Portal Revamp, v1 scope

COMPLETE FEATURE SET
Source: FRD v1.0 Draft
16 Oct 2026 | Prepared by Sadiq

31	12	18	34	43
PUBLIC TEMPLATES	AUTH SCREENS	PORTAL SCREENS	ADMIN SCREENS	MODALS AND DRAWERS

01 | THE FIVE COMPONENTS

What the platform is made of

A — Public website
Visitors, no login. Articles, reports, research, newsroom, Global Watch globe, videos, policies, contact and apply to write.

B — Contributor portal
Article Writers, Research Writers, Experts and Approvers writing. Apply, write, submit, track, revise.

C — Approver portal
Professors and senior analysts peer reviewing research papers double blind, round by round.

D — Admin dashboard
Super Admin and custom roles. Users, content, taxonomy, featured, newsroom, ads, CMS, cards, analytics, logs.

E — ID verification
Anyone scanning a PSF staff ID QR code. Unguessable URL, noindex, active or not valid only.

02 | APPLICATION ZONES

How the single Next.js app is divided

ZONE	ROUTE PREFIX	RENDERING	INDEXING AND ACCESS
Public site	/	Static plus ISR, revalidated on publish	Indexed, open to everyone
Auth	/sign-in, /join, /set-password, /reset-password	Dynamic	noindex
Portal	/portal/*	Dynamic, authenticated	noindex, disallowed in robots.txt
Admin	/admin/*	Dynamic, authenticated, permission gated	noindex, disallowed in robots.txt
Verification	/verify/[token]	Dynamic	noindex, nofollow, never linked or in sitemaps
API	/api/*	Route handlers and server actions	Server side permission check on every call

03 | IN SCOPE FOR V1

Everything the build delivers

- **Public site** of 31 page templates with full CMS control
- **Two track contributor signup** with email OTP verification
- **Invite based accounts** for approvers, experts and staff
- **Dynamic RBAC**, 55 permissions, unlimited custom roles
- **Article, report and research workflows** end to end
- **Double blind peer review** with one or two approvers
- **Newsroom** with home ticker, breaking label and pinning
- **Global Watch globe** plotting every item at its location
- **Featured and top stories** scheduling with holding rule
- **Ad campaign management** with analytics and PDF export
- **CMS** for every public page, draft, preview, publish, rollback
- **Contact, corrections, advertising and copyright inboxes**
- **Activity logs, analytics, ID card verification**
- **Archive migration** 2020 to 2024 with 301 redirects
- **Rich link previews** generated per item for sharing

04 | OUT OF SCOPE FOR V1

Decided and parked

ITEM	DECISION
Urdu or second language	English only, no i18n layer, permanent
Reader accounts	No bookmarks, comments or newsletter. Share buttons only
Online ad payments	Ad bookings are internal only
Paid membership	No gated content, all research open access
Native mobile apps	Responsive web only
Plagiarism detection	Integration hook reserved, provider later
Public team directory	No team on About. Author byline pages instead

Twenty open items remain flagged in the FRD for PSF confirmation, led by the exact logo green hex, the research licence, ISSN and DOI registration, and inbox email addresses.

05 | END TO END

The journey of one piece of content

01 — Apply
A writer picks a track, verifies their email by code and submits an application with expertise, institution and background.

02 — Approved
An admin reviews the application against its submission context and grants the matching roles, or returns it with comments.

03 — Write
The portal editor autosaves every ten seconds, versions every save, and collects classification, location, images, SEO and declarations.

04 — Submit
A full screen preview of the live page and the share card precedes submission. The item locks and gets a reference ID.

05 — Decide
Editors approve, schedule, request changes or reject. Research instead goes to screening and double blind peer review.

06 — Publish
Publishing now or at a scheduled minute, revalidating every page the item touches and generating its share image.

07 — Circulate
The item reaches home slots, its topic and subtopic pages, the globe at its location, search, RSS and the author's own page.

Every requirement in the PSF brief is traced to a numbered section

Appendix E maps all 93 brief requirements to the module that delivers them, from the white and ivory theme through to the rotating coverage globe and the ID card verification page.

93

06 | INFORMATION ARCHITECTURE

Six top level items, all category levels dynamic

Topics

Three level mega menu. Topics column, subtopics with icons and counts, featured item for the hovered topic.

Reports and Research

Dropdowns built from active categories, plus how peer review works and submit a paper.

Global Watch, Newsroom, About

Direct links plus an About dropdown. Newsroom carries a live dot when a post landed in the last two hours.

07 | PUBLIC SITEMAP

Every addressable template

/ Home
/topics hub /topics/[topic] /topics/[topic]/[subtopic]
/articles/[slug] article detail
/reports /reports/category/[category] /reports/[slug]
/research /research/category/[category] /research/papers
/research/[slug] /research/peer-review /research/submit
/newsroom /newsroom/[id]-[slug]
/global-watch /authors/[username] /search
/videos /where-we-are /about
/editorial-policy /corrections /contributor-agreement
/publish-with-us /contact /advertise
/privacy /cookies /disclaimer /copyright /terms
/verify/[token] hidden
/rss.xml /rss/[topic].xml /sitemap.xml /llms.txt
404 | 410 | 500 | Maintenance states

08 | GLOBAL COMPONENTS

Present on every public page

- **Utility bar.** Date in PKT, Global Watch count, reader text size control, social icons, advertise link
- **Header.** Logo lockup, six item nav, mega menu, search overlay, sign in, Write for PSF, bell and avatar when signed in, sticky after 120px
- **Search overlay.** Instant grouped results after two characters, recent searches, keyboard driven
- **Footer.** Five link columns, dynamic topics, social row, automatic copyright year, cookie settings, all CMS editable
- **Cookie consent.** Accept, reject and settings at equal weight. Essential, analytics, social and video categories
- **Share bar.** Seven targets, sticky rail on long form, native share sheet on mobile
- **Rich preview cards.** Dynamic 1200×630 share image per item, with report, research, newsroom and author variants
- **Ad slot component.** Labelled, collapses when empty, viewability counted at 50 percent for one second
- **Breadcrumbs, back to top, 404, 410, 500 and maintenance states**

09 | HOME PAGE ANATOMY

Fourteen sections, admin ordered and toggleable

H1 Hero and ticker

Lead story on a full bleed image with scrim, plus a live newsroom ticker panel of the last 12 posts auto scrolling every five seconds.

H2 Topic quick access

Up to six topic tiles with icon, item count and the first four subtopics as links.

H3 Top stories

One large and three standard, driven by a four position featured slot with most read fallback.

H4 Leaderboard ad

970×250 desktop, 320×100 mobile, booked through the campaign calendar.

H5 Latest coverage

Admin curated top four in a 2×2 grid beside a just published ruled list of the eight newest items of any type.

H6 PSF Coverage globe

Compact rotating globe of the last 90 days with locations covered, items this month and a top locations list.

H7 Topic desks

One block per topic flagged for home, alternating two layouts. Topics under three published items hide themselves.

H8 Reports

Featured report with portrait cover, report number, executive summary excerpt and PDF download, plus two latest.

H9 Research and academia

Three peer reviewed cards plus a publish with PSF panel showing the four step process and review statistics.

H10 Sponsor carousel

Up to five slides, auto advancing every six seconds, pausing on hover.

H11 Most read

Ranked one to five across all content types from the last seven days. Ranking only, no public counts.

H12 Videos

Latest four from the YouTube sync, pinned first, opening in a privacy mode lightbox player.

H13 PSF on social

Live X and Instagram embeds, consent gated and lazy loaded, falling back to follow cards.

H14 Write for PSF band

Deep green band splitting the two contributor tracks with a route into each signup.

CMS control

Section headings, subheadings, visibility toggles, desk order and band copy are all editable without a developer.

10 | LISTING TEMPLATES

How readers browse once they leave the home page

TEMPLATE	SECTIONS IN ORDER
Topic page	Hero with topic image and item count, subtopic tiles with icons, featured item, ad banner, filtered card grid with pagination, sidebar of most read and a coverage map thumbnail
Subtopic page	The same frame with the subtopic icon, a sibling chip row, its own featured slot and a listing restricted to the subtopic
Topics hub	One block per active topic with description, subtopic tiles and three latest items, plus a coverage statistics strip
Reports and research hubs	Hero, featured item, category tiles, filtered grid. Research adds the publishing process, standards, review statistics and an optional call for papers
Author page	Avatar, designation, badges, bio, expertise chips, external links, counts by type and tabbed work
Search	Facets, sort, highlighted terms, snippets and a zero result state that routes into topics

Filters available on listings

Subtopic | Content type | Content label | Report category | Research category | Paper type | Author | Institution | Country | Region | Year | Date range | Tag | Sort by latest | Sort by most read | Sort by relevance

- Filter state lives in the URL, so any view can be shared or bookmarked
- Filtered pages canonicalise back to the unfiltered URL for search engines
- New topics, subtopics and categories appear in navigation, filters and submission forms automatically once set active
- A category holding content can never be deleted, only deactivated or merged, and merging leaves 301 redirects behind

Public website, part two: content, coverage and policy

11 | THE FOUR CONTENT TYPES

What can be published and what each carries

Type 01
Article

Analysis, explainer, opinion, interview, investigation, profile, book review.

- Label, topic and subtopic, location chip
- Standfirst, byline, read time, updated note
- Key takeaways box and footnotes
- Pull quotes, galleries, tables, embeds
- Corrections note top and bottom
- Author box, tags, related coverage
- Archive banner for migrated items

Type 02
Report

Everything an article carries, plus the report apparatus.

- Report number, category, page count
- Executive summary box
- Three to seven key findings
- Auto table of contents, sticky
- Full report PDF download, counted
- Numbered figures with source lines
- Citation modal, five formats

Type 03
Research paper

Peer reviewed, open access, no advertising on the page.

- Peer reviewed badge and paper type
- Authors with institution and ORCID
- Received, accepted and published dates
- DOI, volume, issue and ISSN
- Abstract, keywords, references
- Declarations and supplementary files
- Licence and peer review statement

Type 04
Newsroom post

Short, fast, admin written. Feeds the ticker and the globe.

- Headline, summary with links
- Region, country and city, always
- Breaking label, auto off after six hours
- Pin to top, maximum three
- Optional thumbnail and source link
- Schedule and expiry window
- Shareable permalink with rich card

12 | GLOBAL WATCH

Coverage plotted on a rotating globe

PSF Coverage layer

- One point per location with published content
- Point size scales with item count, four steps
- Point colour shows recency, breaking pulses red
- Hover gives counts by type and latest headline
- Click flies the camera and opens the drawer

Status layer

- Admin marks countries and regions
- Active conflict · High tension · Watch
- Note, start date and review by date
- Expired reviews surface in the action centre
- Visitor toggleable, on by default

- **Filter panel.** Content type, topic and subtopic, date presets, region, country search that flies the globe, layer toggles
- **Location drawer.** Counts by type, tabbed item lists, see all coverage in country, shareable deep link
- **Below the globe.** Region tabs with country chips, filtered card grid, top ten most covered countries, recent activity timeline
- **Accessibility.** A view as list mode renders the same data as a keyboard and screen reader friendly table
- **Performance.** Lazy loaded after main content, 350KB budget, static nightly snapshot as the fallback image

13 | INSTITUTIONAL PAGES

Credibility, policy and conversion

PAGE	WHAT IT CARRIES
About PSF	Mission, who we are, four pillars, live desks, by the numbers, 2019 to present timeline, independence statement. No team listing
Editorial Policy	Thirteen clauses from independence and sourcing through AI disclosure and archive handling
Corrections	Policy, request form and a public corrections log
Contributor Agreement	Fourteen clauses. Accepted at signup and at every submission
Publish With Us	Two tracks, what we look for, formats and word lengths, process, FAQs
Contact	Routing guide and form. No address or phone shown
Advertise	Sales layout. Audience in numbers, reader cards, placement mockups, formats, process, enquiry form
Where We Are	Official channel cards with handles and an impersonation note
Videos	Pinned feature, playlist tabs, synced grid, lightbox player
Author pages	Byline profile, badges, expertise, links, stats and tabbed work
Legal set	Privacy, cookies with a live cookie table, disclaimer, copyright and takedown form, terms

14 | TAXONOMY AND LOCATIONS

Everything classifiable, nothing hardcoded

Topics and subtopics

Two levels, drag and drop ordered, each with slug, description, hero, icon and its own visibility flags.

Defence · Land · Air · Sea · Strategic Systems · Cyber and Information · Geopolitics · South Asia · Afghanistan · Americas · Middle East · National Affairs · Security · Economy · Policy · History · Battles and Operations · Profiles

Section categories

Reports and research carry their own category trees, created and reordered by admins at any time.

- Reports: strategic assessments, special reports, annual outlooks, event readouts
- Research: research papers, policy briefs, emerging scholars

Content labels

A fixed editorial vocabulary so readers always know what they are reading.

Analysis · Explainer · Opinion · Interview · Investigation · Profile · Book Review · News Analysis · News · Report · Research Paper · Policy Brief · Archive · Breaking

Locations

A standard open dataset of countries and cities with coordinates, around 25,000 cities and the full Pakistan list.

- Ten PSF analytical regions, each country mapped to exactly one
- Cascading region, country then city picker everywhere
- A country wide option for items without a city

15 | DESIGN FOUNDATIONS

Institutional, not decorative

Colour

White and ivory surfaces, logo green as the single accent, near black ink. No yellow. Semantic red, amber brown and blue for states, with a fixed status chip set.

Typography

Source Serif 4 for editorial, Inter for interface. Thirteen token type scale, 68 character measure, tabular numerals, reader text size control.

Components

Twenty eight specified components with every state, from buttons and cascading location pickers to tables, drawers, steppers and KPI tiles.

Must not look AI generated

A binding do not list: no gradients, glassmorphism, pill cards, emoji icons, stock weapon art, centred hero taglines, filler statistics or animated counters.

Accounts, contributor portal and peer review

16 | HOW ACCOUNTS COME INTO BEING

Self signup for writers, invite only for everyone else

STEP 1
Choose a track
Write articles and reports, or publish research. Approvers and experts cannot sign up here.

STEP 2
Account details
Name, email, password, expertise from live topics, institution, designation, optional photo, agreements and bot check.

STEP 3
Email OTP
Six digit code, ten minute expiry, five attempts, resend after sixty seconds. No user record exists until this passes.

STEP 4
About you
Research track collects programme, graduation status, registration number, optional ID, ORCID and interests. Article track collects background and samples.

STEP 5
Admin decision
Approve, reject with comments for resubmission, or reject permanently and blocklist the email. Three to five working days.

Invite flow
Admin creates the user, a single use link valid 72 hours sets the password, mandatory two factor setup for admin roles and approvers, then sign in. Resend and revoke available.

Account security
Argon2id hashing, breached password check, lockout after five failures, TOTP with ten backup codes, active session list with remote sign out, email change by OTP.

Status routing
Pending, returned, invited, deactivated and banned accounts each land on their own screen with their own message. Eleven account states in total.

17 | CONTRIBUTOR PORTAL

Write, submit, track and revise

Dashboard
- Six KPI tiles including total reads and 30 day delta
- Daily reads chart, 30 or 90 days
- Top five published items by reads
- Research status mini kanban
- Action needed list and recent activity
- Where your readers are, top five countries

My content
- Tabbed lists with counts for articles, reports and research
- Live, in review, drafts, changes requested, rejected, scheduled
- Withdraw, duplicate, delete and view on site
- Request update on live items, routed to editors
- Filters by topic, status and date, sortable by reads

The editor
A main writing column with a settings sidebar of six collapsible panels: classification, location, images, authors, SEO and declarations.

Full rich text toolbar · Footnotes · Pull quotes · Callout boxes · Tables · Galleries · Full width infographics · YouTube and X embeds · Internal link search · Clean paste from Word · **Autosave every 10 seconds** · **Version history and restore** · Live word count and read time · Co-authors up to four · Live SEO and share preview · **Full screen preview before submit** · Reference ID on submission

18 | RESEARCH SUBMISSION

Three steps and a declaration gate

STEP	COLLECTED
1 Paper details	Title, paper type, category, topic, locations, abstract, keywords, co-authors including external ones, corresponding author, funding, conflicts, ethics, data availability
2 Manuscript	Full text with numbered sections and figures, structured references in APA or Chicago, auto figure and table list, word limits per paper type
3 Files and declarations	Anonymised manuscript PDF with an automatic metadata check, title page, supplementary files, images, SEO, six declarations, suggested and excluded reviewers

Author visibility after submission
- A stage timeline with dates for every transition
- Decision letters, with reviewers shown only as Reviewer 1 and Reviewer 2
- Every version downloadable, round by round
- Revise and resubmit with a required response to reviewers
- Withdraw with a reason, until the final decision

19 | PEER REVIEW

Double blind, with the system enforcing anonymity

1
Screening
Admin checks scope, anonymisation and formatting within seven days. Desk reject, return for formatting, or send to review.

2
Assignment
One or two approvers matched on expertise, with load and turnaround shown. Conflict check blocks self review.

3
Review
Accept or decline with reason, 21 day default window, reminders at seven days, two days and on the due date.

4
Decision
Approve, minor or major revisions, or reject. Conflicting recommendations escalate to an editor decision.

5
Publication
Final approval sets date, DOI, volume, issue and licence. Retraction is available afterwards to Super Admin.

- **Review workspace.** Split screen, rendered paper and PDF viewer on the left, review form on the right, with highlight to comment anchored to passages
- **Round two.** Previous version, text diff, the author response and the reviewer's own earlier comments are all on hand

What a review captures
- Recommendation, four options
- Five scores from one to five: originality, rigour, evidence, clarity, relevance
- Comments to author, minimum length enforced
- Confidential comments to the editor
- Optional annotated PDF

Anonymity rules
- Approvers never see author names, emails, institutions or the title page
- Authors never see approver identities
- Reviewers never see each other
- Enforced in the API, not the UI, and covered by automated tests

20 | LIFECYCLE STATES

Every status the system can hold, and nothing in between

Account, eleven states
Pending signup · Application draft · **Pending** · **Returned** · **Invited** · **Active** · Deactivated · Withdrawn · Invite revoked · **Banned** · **Permanently rejected**

Banned and permanently rejected are terminal. The last active Super Admin can never be demoted or removed.

Article and report, nine states
Draft · **In review** · **Changes requested** · **Scheduled** · **Published** · Unpublished · **Rejected** · **Removed on ban** · Deleted

Research paper, sixteen states
Draft · **Submitted** · **Returned for formatting** · **Desk rejected** · **Under review** · **Editor decision** · **Revisions requested** · **Resubmitted** · **Awaiting final approval** · Minor edits · **Scheduled** · **Published** · **Rejected** · Withdrawn · **Retracted** · Unpublished

Admin dashboard: twenty modules, all permission gated

21 | ADMIN MODULES

Twenty areas, each visible only to the roles that hold its permissions

01 Overview
Action centre with eleven counters, traffic KPIs for today, 30 day visitor chart, hourly heatmap, geography, content KPIs and a live now panel.

02 Analytics
Six tabs: audience, content, acquisition, geography, contributors, engagement. Compare to previous period, CSV and chart export.

03 Users
One table, ten pre filtered views. Roles, status, published and pending counts, last active, signup country, bulk actions.

04 User detail
Profile, full application as submitted, security and sessions, submissions, review history, activity log and internal notes.

05 Create user
Role first stepper, fields adapting to the roles chosen, then invite by email or a copied link.

06 Applications
Oldest first queue, secure ID viewer, submission context with duplicate and domain match flags, three decision paths.

07 Roles
Custom roles from 55 grouped permissions, live plain English summary, duplicate, and safe delete with reassignment.

08 Articles
Full listing with fourteen filters, hover cards, version diffs, link and alt text checks, and a six action decision bar.

09 Reports
Same workflow, separate queue. Report number assignment, PDF replacement with versioning, findings and summary editing.

10 Research
Screening, assignment, review tracker per approver, editor decision, final approval, correction and retraction.

11 Newsroom
Fast table with modal editing, pinning, breaking flag, scheduling, expiry and click counts.

12 Featured
Eleven slot families across every page, a Gantt timeline per position and a six step placement wizard.

13 Taxonomy
Topics and subtopics as a drag and drop tree, plus report categories, research categories and tag merging.

14 Global Watch
Status layer entries, globe defaults and a coverage health view listing items still missing coordinates.

15 Ad campaigns
Campaigns, placements with conflict free booking, a slot calendar, live analytics and a branded PDF report.

16 Inbox
Unified queue for contact, corrections, advertising and copyright, with assignment, notes, reply and spam handling.

17 ID cards
Create a card ID and designation, get a unique URL and downloadable QR, activate or deactivate. Never deleted.

18 CMS
Every public page in locked templates, with draft, preview, publish, revision history and rollback.

19 Activity logs
Append only record of every action with actor, target, IP, device, result and a before and after diff.

20 Settings
General, email templates, integrations, security policy, data retention, maintenance mode and cookie version.

22 | ROLES AND PERMISSIONS

Fifty five capabilities, bundled freely

SYSTEM ROLE	CREATED BY	HOLDS
Super Admin	Seed	Everything, including roles, settings and logs. The last one cannot be removed
Admin template	Super Admin	Editorial and platform management, without roles, settings or log export
Article Writer	Signup or invite	Write and submit articles and reports
Research Writer	Signup or invite	Write and submit research papers and reports
Approver	Invite only	Peer review assigned papers, and write as a contributor
Expert	Invite only	Write articles and reports, carries a PSF Expert badge

- A user may hold several roles at once and their permissions are the union of them
- Custom roles such as Editor, Research Editor, Ad Manager, Analyst Viewer and Moderator are built from the same checklist
- Every check runs server side on every request. Hiding a menu item is never the only control
- Nobody can change their own roles, and only a Super Admin can grant role management

23 | PLACEMENT ENGINE

Featured slots and advertising

Featured and top stories
- Eleven slot families covering home, topics, subtopics, reports, research and the mega menu
- Slots are created automatically whenever a new category is created
- One active placement per position, with conflicts surfaced and a replace option
- **Holding rule.** An expired item stays visible until something replaces it, and the action centre flags it
- Every fallback is defined, so no position is ever empty

Advertising
- Eight slots across home, topics, articles, newsroom, reports and Global Watch. None on research, author, policy or portal pages
- Booking by slot, date and time, with occupied windows disabled in the picker
- Exact pixel validation on upload, desktop and mobile creatives, carousel up to five slides
- Impressions, clicks, CTR, device, country and hour, exported as a branded PDF with the campaign ID

24 | RUNNING THE SITE WITHOUT A DEVELOPER

What PSF controls directly after handover

Content management
Every public page sits in a locked template with typed fields. Edits save as drafts, preview on a signed URL, publish with one click and keep a full revision history with rollback.

Navigation and brand
Menu items, footer columns, social channels, logo, favicon, default share image, contact addresses and the borders disclaimer are all editable in settings.

Media library
Every uploaded asset with search, type and usage filters, dimensions, alt text and credit, replace keeping references intact, and deletion blocked while in use.

Policy and retention
Password rules, session lengths, two factor enforcement, lockout thresholds, IP blocklists, retention windows and maintenance mode, with a cookie version bump to re prompt consent.

Nothing happens on the platform without a record of it
Sixteen action groups are logged append only, from sign ins and bans to permission diffs, featured transitions, CMS publishes and settings changes. Page views and reads stay in analytics so the log remains readable.

16

Platform layer and delivery

25 | CROSS CUTTING SYSTEMS

The machinery behind every module

Measurement

- A read counts after ten seconds or a quarter scroll, deduplicated per IP per item per 24 hours
- Bots, prefetches, previews and self views excluded
- Twelve tracked events including shares, PDF downloads, globe clicks and ad impressions
- Daily salted hashing for unique visitors, geo at ingestion
- Raw events for thirteen months, aggregates kept indefinitely

Discoverability

- Per item SEO fields with a live search preview
- Eight JSON-LD schema types including ScholarlyArticle
- Highwire Press tags so research reaches Google Scholar
- Sitemap index per type plus a Google News sitemap
- Site wide and per topic RSS, llms.txt regenerated nightly
- A managed redirect table, editable without a deploy

Notifications

- Thirty four triggers across email and an in app bell
- One branded template set, all editable in settings
- Security and account mail cannot be switched off
- Admins choose immediate or an 08:00 PKT digest
- Bounce and complaint webhooks flag dead addresses

Security and privacy

- Mandatory two factor for admin permissions and approvers
- Rate limits on sign in, OTP, forms, search, tracking and verification
- Pre signed uploads, private bucket for sensitive files, malware scanning
- Allowlist sanitised rich text, CSP with nonces, restricted embeds
- ID documents deleted 30 days after a decision, raw IPs anonymised after 90
- Point in time recovery, daily encrypted backups, quarterly restore test

Accessibility and performance

- WCAG 2.1 AA across public site, portals and admin
- Full keyboard operation of the mega menu, editor and globe
- Mandatory alt text on every upload, enforced by the CMS
- LCP under 2.5 seconds, INP under 200ms, CLS under 0.1
- Lighthouse CI in staging blocks a failing production deploy
- 99.9 percent monthly availability target with minute by minute uptime checks

Integrations

- YouTube Data API, uploads and playlists synced every six hours
- X and Instagram embeds, loaded only after consent
- MaxMind GeoLite2, refreshed weekly for geography
- Resend for transactional email with full domain authentication
- Cloudflare Turnstile on signup and every public form
- A reserved similarity check interface for plagiarism, later

26 | UPLOAD GOVERNANCE

Twenty five upload fields, each with its rules shown before a file is chosen

Recommended size shown on every field | Fixed ratio croppers | Client and server validation | EXIF and GPS stripped | WebP and AVIF conversion | Six responsive widths | Malware scan on documents

Private bucket for IDs and manuscripts | Signed URLs valid ten minutes | Drag and drop with progress | Replace keeps references intact

27 | STACK AND BUILD ORDER

One codebase, seven phases

LAYER	CHOICE
Framework	Next.js App Router with TypeScript, front end and API in one codebase
Hosting	Vercel, preview deployment per branch, three environments
Database	PostgreSQL with Prisma, full text search and trigram extensions
Auth	Auth.js with database sessions, argon2id and TOTP
Storage	Cloudflare R2, public and private buckets behind a CDN
Editor and globe	TipTap stored as JSON, react-globe.gl on Natural Earth polygons
Supporting	Upstash Redis, Resend, Turnstile, Recharts, react-pdf, Sentry, Vercel Cron

PHASE	SCOPE	EXIT CRITERIA
0	Foundations	Tokens, components, schema, auth skeleton, uploads and email in staging
1	Identity and access	Every signup, application, invite and RBAC flow passes acceptance tests
2	Content core	An article runs from signup to published with working share previews
3	Research	A two approver paper completes end to end with identities provably hidden
4	Home and discovery	Featured holding rule verified, globe inside its performance budget
5	Platform	All CMS pages editable, ad overlap prevention tested
6	Migration and launch	Archive migrated, zero 404s on old URLs, launch checklist complete

28 | LAUNCH GATE

What has to be true before the site goes live

- Logo, exact green hex and brand assets in place
- All CMS pages populated, legal pages reviewed by a lawyer
- Seed taxonomy, regions and locations loaded with heroes and icons
- Archive migrated and every old sitemap URL redirecting
- Super Admin accounts created with two factor, custom roles defined
- Email domain authenticated and every template tested
- Turnstile, GeoIP, YouTube, X and Instagram configured
- Cookie consent tested across accept, reject, settings and re prompt
- Share previews verified on WhatsApp, Facebook, X, LinkedIn, Telegram
- Double blind anonymity tests passing
- Featured holding, ad overlap and scheduled publishing tested
- Core Web Vitals met on home, article, topic and Global Watch
- WCAG 2.1 AA audit passed, automated and manual
- Security headers, rate limits, backups and a restore test done
- Monitoring, alerting and uptime checks live
- robots.txt, sitemaps, llms.txt and security.txt live
- Staging confirmed noindex and password protected
- Admin training session held and documentation handed over

Archive migration carries 2020 to 2024 across without losing a URL

Content converted to the new editor format, original dates and bylines preserved, images rehosted, locations tagged with an assisted pass so archive items join the globe, and every legacy address redirected and verified by an automated crawl.

301